

Exercise 24

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Introduction to Electrodynamics

ISBN: 9780321856562

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From the Problem 2.16) we know that the electric field in space is:

$$\vec{E} = \begin{cases} \frac{\rho s}{2\epsilon_0} \hat{s} & , s \in [0, a] \\ \frac{\rho a^2}{2\epsilon_0 s} \hat{s} & , s \in [a, b] \\ 0 & \text{if } s > b \end{cases}$$

The potential is then easily calculated by integration:

$$\begin{aligned} V(s) &= - \int_0^b E ds = - \int_a^b \frac{\rho a^2}{2\epsilon_0 s} ds - \int_0^a \frac{\rho s}{2\epsilon_0} ds \\ &= - \frac{\rho a^2}{2\epsilon} \ln \frac{b}{a} - \frac{\rho}{2\epsilon_0} \frac{1}{2} a^2 \\ &= \boxed{- \frac{\rho a^2}{4\epsilon_0} \left[1 + 2 \ln \frac{b}{a} \right]} \end{aligned}$$

Result

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$$\boxed{\Delta V = - \frac{\rho a^2}{4\epsilon_0} \left[1 + 2 \ln \frac{b}{a} \right]}$$

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